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REVIEW ARTICLE

Hazards threatening underground transport systems

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Abstract
Metro systems perform a significant function for millions of ridership worldwide as urban passengers rely on a secure, reliable, and accessible underground transportation way for their regular conveyance. However, hazards can restrict normal metro service and plans to develop or improve metro systems set aside some way to cope with these hazards. This paper presents a summary of the potential hazards to underground transportation systems worldwide, identifying a knowledge gap on the understanding of water-related impacts on metro networks. This is due to the frequency and scope of geotechnical and air quality hazards, which exceed in extreme magnitude the extreme precipitation events that can influence underground transportation systems. Thus, we emphasize the importance of studying the water-related hazards in metro systems to fill the gaps in this topic.

Keywords Urban climate adaptation · Hazards assessment · Critical infrastructure networks · Metro system · Subway

1 Introduction

The globalization process has grown these last 60 years, considering as Africa and Asia are urbanizing quicker than the rest of the continents in the coming decades (UN 2018). As shown in Fig. 1, projections indicate a growth of the world's urban population by more than two-thirds by 2050, with almost 90% of that increase in urban areas of Asia and Africa.

A wide range of environmental hazards including extreme weather events, droughts, biodiversity loss, and stress on natural resources are impacting on cities worldwide. The most significant hazards since 2011 concerning probability and global impact are the extreme weather events and the lack of adaptation to climate change.

Therefore, current trendlines involve encouraging shifts in priorities at the governmental and private levels and focus on vulnerable growing cities to the impact of climate change (WEF and Collins 2019).

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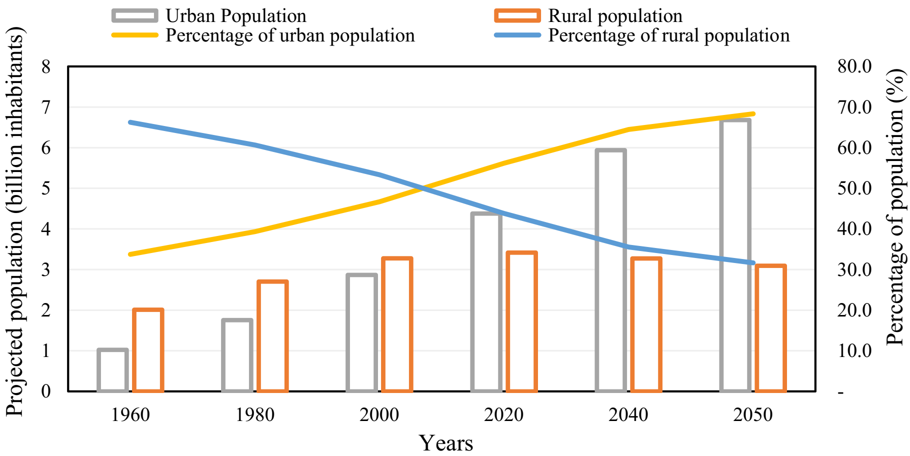

Fig. 1 Historical and projected evolution of the urban population compared to the world's rural population, 1960–2050 (UN 2018)

Urban areas can be considered as living organisms, comprising several interdependent sectors and activities, intimately connected as services. Climate change requires cities to mitigate several hazards in the short term, yet they must develop their potential to improve their resilience (Kim and Lim 2016).

To achieve cities sustainability, we need to analyse urban resilience. Given the growing people and resources concentration on urban environments, as the increasing frequency and intensity of risks that threaten their services, the cities life cycle should be examined (Sharifi and Yamagata 2018). Urban resilience, as a city system recovery potential facing different hazards, becomes relevant taking into account service interruptions worsened by climate change next century (Velasco et al. 2018).

The interdependence linking the city services, such as water, energy, and public transport as critical infrastructure networks, has increased due to technological advances in recent decades. This bond has generated incremental improvements in essential city services quality and coverage, but a worsened status when a natural hazard event occurs impacting the service operational viability. Subsequently, the failure across the network of critical infrastructure services spreads, known as cascade effects (Evans et al. 2018).

One of the most critical services for the proper functioning of a city is public transport networks. According to the International Association of Public Transport (UITP), the year 2015 noticed an 18% increase in public transport trips compared to 2000, with 243 billion trips made in 39 countries (UITP and Saeidizand 2015).

The backbone of urban mobility is well-integrated high-capacity public transport systems into a multimodal arrangement. In both developing and developed countries, most maintain or increase the market share of formal public transport (United Nations Human Settlements Programme 2013).

Due to the increase in population and awareness to achieve a lower economic, environmental, and social impact, cities worldwide are implementing public and non-motorized transport systems as metro networks. Sustainable transport systems have a positive correlation with GDP, while vehicle use improves economic and social parameters, albeit with a negative impact on the urban environment (Haghshenas and Vaziri 2012).

The accelerated metro systems development since the 1960s responding to mega-cities growth shows their importance holding public mobility in urban areas. Metro infrastructure is less bottlenecks-prone than roadways and mitigates long distances to urban activity nodes for the population living in peripheral locations (United Nations Human Settlements Programme 2013).

According to the cities stimulated growth, consequently, of their metro systems, the higher the growth and metro network complexity, the higher the vulnerability to natural hazards (Sun and Guan 2016).

Metro systems concentrate the corridors with the highest volume/length of travel and the greatest activity centres in the cities (Yang et al. 2015). Underground transport systems are an essential part of the urban lifestyle, as passengers rely on a safe, reliable, and accessible system for their regular transportation (Mohammadi et al. 2019).

A variety of definitions of the term “metro” are accepted such as subway, underground, or tube. Throughout this document, the term “metro system” is used to refer to high-capacity underground urban railway systems, which are operated under an exclusive right of way, using the definition suggested by UITP (2018).

In its report, World Metro Figures 2018, UITP (2018), summarizes some facts of urban underground transport systems. They operate in 178 cities and 56 countries by 2018, carrying 168 million passengers per day on average with a 19.5% annual ridership increase worldwide.

However, to date, no conclusive research is known on metro systems general data, such as typology (size, configuration, passengers’ number, depth, length), tunnels and infrastructure administrative, economic, and physical sustainability. Or even, a hazards summary threatens metro systems as important issues; most studies in underground transport systems have just focused on confined conditions.

This document provides a comprehensive and systematic review of studies on risk assessment for underground transport systems, exploring the impact sources and existing assessment methodologies.

The general structure of this literature review has five sections: Introduction presents how underground transport systems such as metro respond to urban population growth dynamics and natural hazards they face, highlighting the importance of this research.

The second part presents the methodology implemented for this literature review. Section 3 introduces risk and resilience in transport systems concepts and the link within facing natural hazards and developing urban resilience. Section 4 begins by laying out the knowledge dimensions of threat categorization research for metro systems. The closing section examines the results of the literature review.

2 Methodology

The methodological approach adopted in this article is a mixed methodology based on the work of the PRISMA Group (Moher et al. 2009a), whose effort provides a reliable method for performing a literature review and is used in similar studies, such as Eckhardt et al. (2019). Figure 2 provides the summary of the performed approach.

Literature review first step establishes the relevance of urban transport systems as a key component, offering a proper integrated operation. Due to the lack of information of natural hazard adaptation measures in metro systems, a significant part of the available

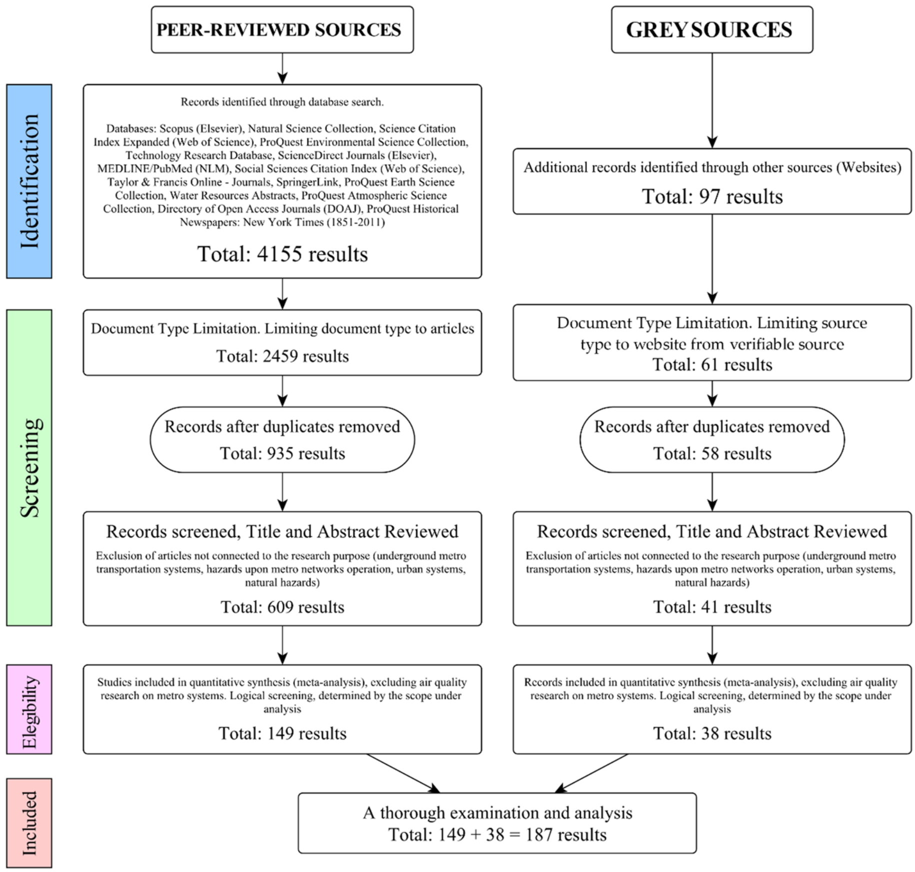

Fig. 2 Literature review output summary. Adapted from the PRISMA Statement (Moher et al. 2009b)

information comes from sources other than academic publications and distribution channels (e.g. newspapers and reports) mainly known as grey literature.

Collecting, organizing, and analysing information processes apply relevant keywords determination to avoiding bias, similar but unrelated sources to the under investigation topic, and other source limitations. Table 1 presents this information along with the eligibility criteria of the data sources.

3 Access to vulnerability and resilience concepts in underground transport systems

Due to practical constraints, this paper cannot provide a comprehensive review of the risk and resilience concepts in transportation systems, and this study has only considered the context of risk and resilience concepts in underground transportation systems.

In a comprehensive literature outline of resilience concept, Wan et al. (2018) identified how growing complexity and unpredictability in transport schemes expose systems to disruptions and risks, varying from natural hazards, such as earthquakes, sea-level rising, and

Table 1 Eligibility criteria for research components

Research component	Criteria establishment procedures	Definition
Research keywords	Description on the keywords generally used to describe hazardous events provoked by nature	"hazard" in any field
	Definition according to the inclusion of synonyms or other words applied to describe underground transport systems	"subway", OR "underground", OR "metro" in any field
	Exclusion of results related to anthropogenic hazards, such as terrorism	NOT "terrorism"
	Exclusion of the results related to the construction stage of metro systems	NOT "construction"—for instance, excavation of urban subway tunnel
Peer-reviewed sources—grey sources	Research that examines the hazard impact assessments in metro-type urban underground transport systems	Studies delimited to the definition of metro as an underground urban railway transport system different from suburban trains or tram systems
	Documents not related to the research purpose	Health and safety hazards associated with the metro system
		Noise levels associated with the metro system
		Generalization of hazards in metropolitan areas
		Simulations of evacuation behaviour during a disaster in the metro system

extreme storms, to critical anthropogenic events such as terrorist attacks and strikes. Also, it defined a summary of discussions and interpretation of terms linked to resilience.

Several studies, in particular Sun and Guan (2016), discuss the exposure of the metro system operation and summarize the different methods for metro vulnerability assessment. The graphical network theory is the preferred method to perform theses analysis, taking into account the specific topological conditions of the metro system such as passenger flows, length, and station capacity, with dynamic traffic redistribution after any failure or attack (Xing et al. 2017).

One of the most used approaches to assess the vulnerability of underground transport services is the service interruptions effects simulation, besides the evaluation of the system-critical elements under demanding conditions (Rodríguez-Núñez and García-Palomares 2014).These studies outline a critical role for passenger flow as a key factor in assessing metro systems vulnerability.

In the same vein, Mattsson and Jenelius (2015) are interested in issues related to a better risk description, such as “a scenario description, the probability, and the consequences (a measure of damage) of that scenario” in transportation system operations. This view goes beyond the traditional description of risk as the product of probability and consequence.

Resilience and risk curve generation complexity are highlighted due to unreliability and vulnerability estimation differences, as risk probability functions in transportation systems. In recent years, available methods have attempted to identify critical nodes in metro networks when assessing the system disruptions impact (M’Cleod et al. 2017).

On the other hand, multiple studies have compared many approaches that evaluate the resilience of current city public transport systems because of their critical importance. Approaches such as that established for the London Underground (D’Lima and Medda 2015) relate the time it may take for the system to recover. However, such approaches do not consider the complexities of natural hazards such as an extreme rainfall event.

As Zhang et al. (2018) concluded, the studies reviewed set a general framework to create a metro system resilience analysis, which studies the network stations connectivity and recovery procedures after network disruptions.

Nevertheless, such studies remain limited in their approach that deals with resolving transport network disruptions. Considering the current and future interdependencies linking the several city services, cascading effects generated by metro system disruptions can affect diverse urban services, indicating needs for additional research to evaluate integrated urban resilience such as the European Project “RESCCUE” (Velasco et al. 2018).

Table 2 summarizes the review of the literature on the components of resilience and vulnerability. As concepts widely implemented in various contexts, this research only covers these notions for metro systems.

4 An overview of hazards categorization for metro systems

Figure 3 shows the identified hot spots and trend lines of current research on natural hazards and vulnerabilities in urban metro transport systems. This research uses the CiteSpace visualization and bibliographic analysis software (Chen et al. 2010) for two purposes. First, it generates an accurate picture, with their importance, of the approaches applied in metro systems natural hazards and vulnerabilities research, and second, it identifies current research potential gaps.

Table 2 Literature review on metro systems resilience and vulnerability components

Source classification criterion	Summary	Source
State-of-the-art review on transport system resilience	This paper introduces a systematic review of transport resilience with an accent on its descriptions, features, and analysis techniques employed in several transport operations. It identifies how reliable transport plays an essential role as a central part of global activities	Wan et al. (2018)
	Within the framework of the "RESOLUTE" project, funded by the EU, this paper examines the methodologies and applications of resilience management for transport systems in several countries, comparing and analysing the impacts of disturbances	Gattinoni and Scesi (2017)
	This paper discusses vulnerability and resilience definitions with related concepts, recognising two diverse ways to study the topic, first, studying transport vulnerability through graph theory, second, demand and supply representation sides. It identifies how short is literature on transport resilience, concerning the response and recovery periods after a failure	Mattsson and Jenelius (2015)
	This paper suggests a comprehensive conceptual framework aimed at expanding the network resilience concept within transport safety at different scales	Reggiani (2013)
Resilience-associated stochastic metrics	This paper introduces a resilience measure by presenting a systems' recovery quantification speed from disruptions, employing a mean-reverting stochastic model to analyse the interruptions diffusive effects and implement this model to London Underground case	D'Lima and Medda (2015)
Step-by-step algorithm for resilience estimation	This paper aims to estimate metro network vulnerability studying disruption from line operation viewpoint using the Shanghai metro network as case research. Results present recommendations on metro system administration for an operational performance potential increase and ridership having an enhanced alternative system when a disruption befalls	Sun and Guan (2016)

Table 2 (continued)

Source classification criterion	Summary	Source
Grid-based (or node) vulnerability analysis	This paper proposes a network model for the New York City subway system with a strategy based on passenger flow simulations on the shortest path to quantify the setbacks suffered by passengers that appear because of disturbing events, mainly those that occur simultaneously, determining separate disturbance scenarios and their results	M'Cleod et al. (2017)
	This paper develops a methodology for estimating public transport network vulnerability, applied to the Madrid Metro system. The study involves disruption consequences in riding times or trips number lost for the entire system with a complete GIS exploration approach. Results show critical links which have low line density and the high ridership number, noticing the circular line importance as a network robustness factor	Rodríguez-Núñez and García-Palomares (2014)
Resilience in response to terrorist attacks	This paper studies terrorist attacks occurrences against metro systems, aiming to decrease attacks number by lessening the transport systems attractiveness as a target, within the European FP7 project SecureMetro. This paper defines critical systems and recommends enhancements to metro carriages design, to increase emergency management capacity, learning from the experience of London Underground bombings and other emergencies	Bruyelle et al. (2014)
Transport systems vulnerability and resilience to cope with flood hazards, sea-level rise, and sea storm surge	This paper provides an analysis of guided transport systems resilience to flooding hazards through failure mechanisms analysis. Applying operational safety methods and concepts and software design is feasible to anticipate all disruption scenarios and domino effects. This paper provides a vulnerability characterization methodology for guided transport systems facing natural hazards and to associate vulnerability depending on whether the system is in an underground, ground-level, or surface arrangement	Gonzva et al. (2017)
Multi-valued resilience and dependency graph frameworks	This study establishes a graphical interdependency model based on Bayesian network and the Delphi method for dynamic assessment of the factors determining fire conditions, fireproof/intervention measures, and fire consequences outcomes in metro stations. This research proposes insights into a practical examination for emergency decision-making towards fire emergency reduction considering the limited dependence in the fire spread process and includes fireproof/intervention measures	Wu et al. (2018a, b, c)

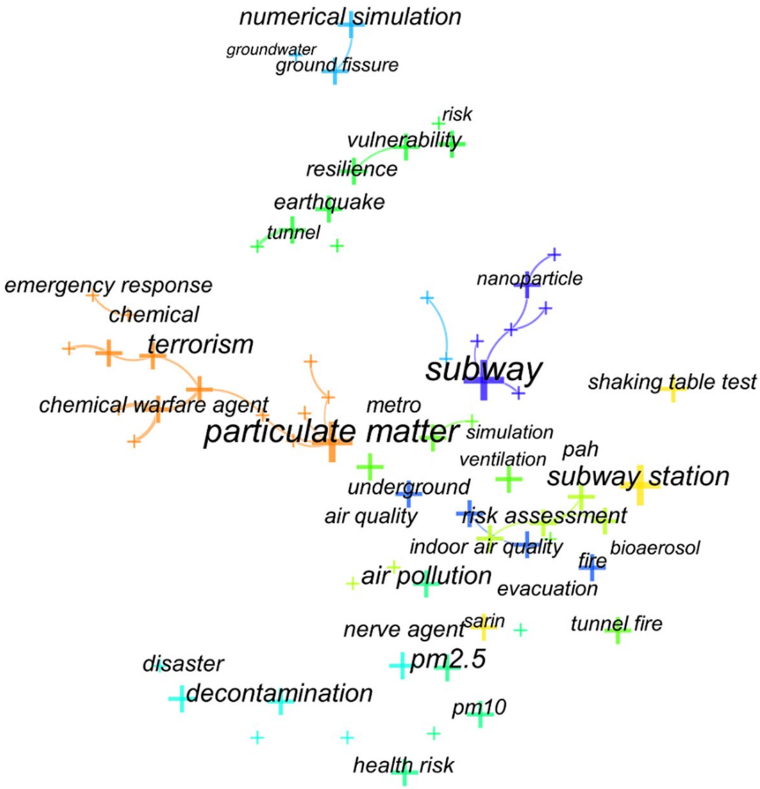

Fig. 3 Visualized networks of co-keywords with highest occurrence frequency. Diagram based on pathfinder network scaling and co-citation analysis theory (Chen et al. 2010)

The application of Table 1 filters to improve the results is not possible in all cases, so some issues appear outside the context of this research (e.g. terrorism) as Fig. 3 shows. Despite these limitations, Fig. 3 represents a comprehensive review of the most important research keywords related to natural hazards on metro systems for current peer-reviewed sources.

Recognizing knowledge groups importance and weight within scientific publications related to this study area is performed by graphical analyses of the keywords association. Figure 3 highlights the predominance of air quality risk research affecting metro systems, showing how the role of particulate matter and air pollution has received increased attention in various research settings in recent years.

Five important research topics emerge from the literature review so far focusing on hazards to metro infrastructure: (a) air quality, as airborne particulate matter; (b) geohazards, expressed by ground fissures and seismic impacts; (c) geohazards, expressed by groundwater flows; (d) water-related hazards such as pluvial or river flooding; and (e) fire risks with smoke management. By far, to date, water-related hazards in metro networks have received limited attention in the research literature.

This document, as metro systems hazard comprehensive review, covers many recent studies focused on metro stations fire hazards. We skim 545 articles in relevant journals between 2009 and 2019. Numerous studies have attempted to explain how to improve air quality in metro systems including detailed reviews of 160 major studies from over 20 countries were thoroughly examined by Xu and Hao (2017). For example, a major fieldwork project on air quality in metro stations was the EU-funded IMPROVE LIFE project (Moreno et al. 2014, 2015a, b, 2018; Martins et al. 2015, 2016; Moreno and de Miguel 2018; Spanish Research Council 2018).

Geological hazards are within the typologies of hazards that may threaten metro systems. Much of the available literature (Dashko 2016; Wu et al. 2018c) deals with planning and construction phases since metro stations settlements during excavations are highly subject to geotechnical problems and the influence of the water table (Raben-Levetzau et al. 2004). As this literature review disregards the metro systems development phase hazards, these research types are not addressing here.

This study identifies a gap in the literature, intending to understand how flooding events in the metro system generate economic and social impacts through metro service disruptions. Reviewing reports of the flood-affected infrastructure in the Tokyo (Ministry of Land 2008), Shanghai (Li et al. 2018a), London (Gonzva et al. 2017), Barcelona (Saurí and Palau-Rof 2017), and New York (MTA New York 2012) systems, it draws attention to considering underground system flood risk assessment as a key factor in an urban resilience analysis.

Table 3 provides an overview of the hazards assessment approaches for metro systems in an organized manner. The hazard classification mentioned at the beginning relates to different studies through a detailed subtopic summary for each hazard category. This summary attempts to highlight the differences between studies focusing on other hazards, extensive, in contrast to the lack of water-related hazards for metro systems.

5 Discussion and future research directions

Resilience concept for public transport systems involves ensuring service availability through operation quality and integrated connectivity with the city transport network. The vulnerability of transportation systems is quantified by the transportation network efficiency when nodes, or in this case, metro stations, suffer service disruptions (Zhang et al. 2018).

Metro systems resilience improvements have focused on examining transport networks efficiency and return times to normal conditions following mathematical models, irrespective of particular risk management and its importance for metro systems resilience improvement, understood as the system's resilience.

Studies such as that conducted by Avci and Ozbulut (2018) present a simplified approach to hazard and vulnerability risk assessment for metro stations and focus on setting the overall assessment for each metro system component, but not on how the various hazards may affect the system as a whole. Although metro systems and stations are diverse, the risks caused by different hazards change in magnitude and importance according to the hazard impacting the metro service.

Decision-makers commit to ensuring the viability of their public transport systems, and that viability entails a priority interest in the system essential operation under normal operating conditions. Geotechnical hazards influence these operational conditions because they involve natural situations such as groundwater intrusion and tunnel fissures, as hazards related to the operation of the system, such as the generation of particulate matter and fires caused by electrical failures.

Table 3 Literature review on hazards affecting metro systems worldwide

Hazard classification	Study approach	Reviewed sources
Airborne particulate matter—air quality		25 papers
	a. Studies of the concentration of particulate matter in tunnels and station platforms at a local level	Cheng et al. (2008), Kam et al. (2011), Querol et al. (2012), Cartenì et al. (2015), Cusack et al. (2015), Perrino et al. (2015), Qiao et al. (2015), Li et al. (2018b) and Cartenì and Cascetta (2018)
	b. Air quality monitoring and prediction studies at metro stations	Kim et al. (2010, 2012, 2017)
	c. Review studies of air quality in underground metro systems	Carteni (2016), Hwang et al. (2017), Xu and Hao (2017) and Moreno et al. (2018)
	d. Studies detailing factors that affect air quality in metro stations	Moreno et al. (2014, 2015a, Martins et al. (2015, 2016) and Li et al. (2018b)
	e. Air quality studies in metro systems carried out in developing countries	Murruni et al. (2009), Mugica-Álvarez et al. (2012)
	f. Numerical models of air quality in metro systems	López González et al. (2014), Qiao et al. (2015) and Moreno et al. (2015a)
Geohazard: Ground fissures and seismic impacts		11 papers
	a. Assessment of the normal stress, shearing stress, or any deformation kind of the section of a metro underground line	Huang et al. (2014 and Shi et al. (2018)
	b. Effects of metro-induced ground-borne vibration	Wu and Xing (2018)
	c. Investigation of the train-induced settlement of a metro tunnel in clays or permeable strata	Di et al. (2016), Huang et al. (2017a) and Tang et al. (2017)
	d. Seismic response of a segmented metro tunnel with flexible joints passing through active ground fissures	Liu et al. (2017)
	e. Geotechnical conditions of deep running metro tunnels	Dashko (2016) and Wu et al. (2018c)
	f. Failure of metro tunnels that pass obliquely through ground fissures at low angles	Peng et al. (2016)
	g. Countermeasures to mitigate the adverse impact caused by the activity of ground fissure	Wang et al. (2016)
Geohazard: Groundwater flows		6 papers

Table 3 (continued)

Hazard classification	Study approach	Reviewed sources
	a. A method used to predict time-dependent groundwater inflow into a metro tunnel	Liu et al. (2018)
	b. Methods used for evaluation of steady-state groundwater inflow to a shallow circular cross-sectional Metro tunnel	Nikvar Hassani et al. (2018)
	c. Impact on aquifers due to the construction of metro tunnels producing changes in the natural groundwater behaviour	Font-Capo et al. (2015)
	d. Groundwater raising or lowering phenomenon modelling due to metro underground infrastructure	Raben-Levetzau et al. (2004), Gattinoni and Scesi (2017) and Colombo et al. (2018)
Fire and smoke		See notes
	a. Ventilation-aided tunnel evacuation systems to create smoke-free evacuation passageway out of the tunnels	Gao et al. (2013) and Liu et al. (2019)
	b. Assessment of the evacuation of passengers in a metro fire event	Zhong et al. (2008), Wang et al. (2013), Lo et al. (2014) and Song et al. (2018)
	c. Risk analysis frameworks for fire safety in underground metro systems	Soons et al. (2006), Wu et al. (2018b)
	d. Infrastructure of vehicles for passengers' life safety facing challenges from fires in metro stations	Li and Dong (2011) and Wang et al. (2018)
	e. Conditions into metro stations during fire events	Gu et al. (2016)
Water-related hazards: Floods due to extreme rainfall or due to river floods		13 papers
	a. Connection linking flood events on the surface with vulnerability to flooding of underground subway infrastructure.	Lyu et al. (2016)
	b. Frameworks based on decision-making methods as networks theory and analytic hierarchy process for assessing the flood evolution process and consequences in underground spaces	Lyu et al. (2018) and Wu et al. (2018a)

Table 3 (continued)

Hazard classification	Study approach	Reviewed sources
	c. Integration of a storm water management model into a geographical information system to evaluate the flood risk in a specific metro system	Herath and Dutta (2004), Li et al. (2018a) and Lyu et al. (2019a)
	d. Methodologies to obtain risk level studying both flood intensity and evacuation difficulty in underground spaces like metro stations	Han et al. (2019)
	e. Analysis of metro systems resilience in the face of flood hazards, studying the components failure steps	Gonzva et al. (2017)
	f. Risk assessment for metro systems flooding events based on regional flood risk evaluation methods	Lyu et al. (2019b)
	g. Evaluation of the waterlogging risk of metro infrastructure caused by rainstorm in a specific metro system	Quan et al. (2011)
	h. Assessment of the risk in a specific metro system against fluvial flooding	Compton et al. (2009)
	i. Evacuation of ridership from inundated underground space	Ishigaki et al. (2008, 2010)

Metro systems hazards classification into five categories by scientific research examines a broad studies spectrum focused on metro stations air quality and the geotechnical hazards that underground infrastructure must control.

Because of the increased frequency and intensity of events associated with these hazards, researchers have focused their efforts on them. Incidents such as fires and the presence of smoke, as physical geotechnical circumstances such as earthquakes, significantly affect ridership of metro's underground networks due to the high prevalence of loss of human life in such events, severe damage to existing infrastructure, and dangerous effects for ridership health.

Advanced metro systems recognize hazard management in their processes thanks to the equipment and infrastructure maintenance, to cope with continuous system variability. Their interest is currently focusing on extreme weather events and hazards due to climate change, considering the increase in high-magnitude events frequency. An influential precedent is Transport for London (TfL), the integrated transport authority who has established action plans for enduring extreme weather events in their metro system (Transport for London 2011).

Researchers have not addressed the water-related hazards in the metro systems in much detail. As Willems et al. (2012) argued, urban flooding and sewerage surcharge due to climate parameters variabilities like extreme rainfall and temperatures, increase urban vulnerability.

Despite the many events, most hazard research is performed by China and Japan, due to massive floods occurrence in metro systems such as Shanghai (Deng et al. 2016; Huang et al. 2017b) and Osaka (Hamaguchi et al. 2016; Terada et al. 2017; Sugimoto et al. 2018) metro systems.

United Nations Global Assessment Report (GAR) 2019 report (UNDRR 2019) indicates urban areas global disasters in 1985 and 2015 were triggered by water-related hazards, except in North America. The UN concludes that localized hazards, including flash floods, urban flooding, and other weather-specific events, are responsible for extensive damage to infrastructure and livelihoods, representing the highest economic losses and impact on development assets such as metro infrastructure (UNDRR 2019).

While hydrological hazards studies are a growing field, to date, relatively little research on floods affecting metro systems exists. The lack of climate change-related hazards studies on metro systems is worrisome.

6 Conclusion

This literature review provides a better understanding of hazards and vulnerabilities in metro systems in four novel ways. First, it presents urban population growth and its intrinsic relationship with hazards in transport systems, focusing on the metro system as city backbone.

Second, it emphasizes the interdependence between public transport services and other services provided in a city, which leads to increased resilience once services are interconnected. Metro systems represent an essential link in urban transport management and as part of the chain, in particular, lack a summary of the potential hazards that can disrupt their operation.

Third, it presents a potential hazards summary that metro systems can experience, categorizing into four classes, identifying one (water-related) as an insufficient studied in-depth hazard type, in comparison with the other three types known.

Fourth, it offers an alternative hazards assessment for metro systems, beyond the conventional view. The traditional approach attempts to improve resilience by decreasing the connection time with another transport node; this research also recommends hazards mitigation, boosting the system's resilience.

As the gap identified in this study, we recognized a lack of scientific information of the water-related hazards affecting metro systems. One of the expected developments from this research is to help inform future developments in water-related hazards as a fundamental component in understanding all the hazards that can affect underground transport systems.

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